Linear Algebra I & Mathematics Tutorial 1b Instructor: Henrik Bachmann
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Quiz 4: Linear maps

Tuesday, 27th October, 2026

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Exercise 2. (4 Points) Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map with

$$F\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}3\\5\\-2\end{pmatrix}, \qquad F\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\4\\-2\end{pmatrix}.$$

Calculate $F\begin{pmatrix}3\\1\end{pmatrix}$.

Solution.

The matrix of F is $A \in \mathbb{R}^{3\times 2}$:

$$\begin{pmatrix}a & b\\ c & d\\ e & f\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}2a+b\\2c+d\\2e+f\end{pmatrix} = \begin{pmatrix}3\\5\\-2\end{pmatrix}, \qquad \begin{pmatrix}a & b\\ c & d\\ e & f\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}a+b\\c+d\\e+f\end{pmatrix} = \begin{pmatrix}1\\4\\-2\end{pmatrix}$$

$$\begin{cases}2a+b=3\\2c+d=5\\2e+f=-2\end{cases} \qquad \begin{cases}a+b=1\\c+d=4\\e+f=-2\end{cases}$$

$$A = \begin{pmatrix}2 & -1\\1 & 3\\0 & -2\end{pmatrix} \implies F\begin{pmatrix}3\\1\end{pmatrix} = \begin{pmatrix}2\cdot 3-1\\3\cdot 1+3\\0\cdot 3-2\end{pmatrix} = \boxed{\begin{pmatrix}5\\6\\-2\end{pmatrix}}$$